Velocity tells us more than speed

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Signed area / ES: área con signo / PT: área com sinal

Above zero adds displacement; below zero subtracts it.

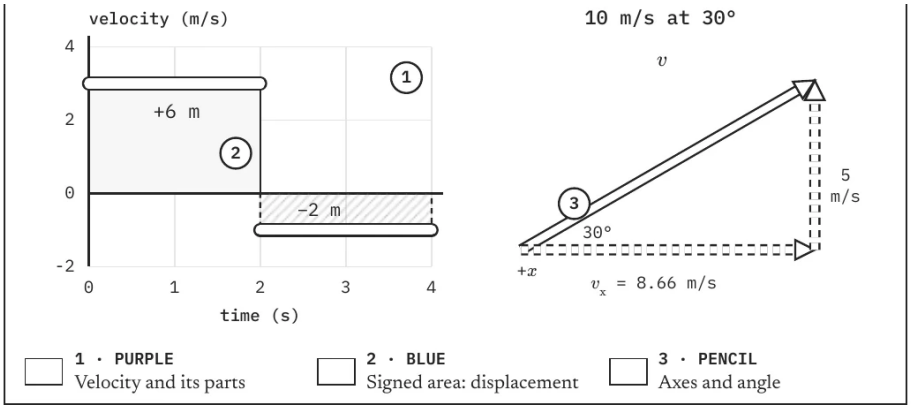
Also on your note page: Distance, Component, Angle θ.

B READ THE RELATIONSHIP

Displacement = signed area under a velocity–time graph

As you read, find: above · below · area.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart moves at +3 m/s for 2 s , then −1 m/s for 2 s . Find its displacement and total distance.

Given v 1 = +3 m/s v 2 = −1 m/s 2 s per interval

Rule Use areas because each interval has steady v .

Work Δ x = 3(2) − 1(2) distance = 6 + 2

Answer +4 m ; 8 m Return travel reduces Δ x , not distance.

E SAVE FOR THE FINAL MEETING

For fictional XJ, the area of a velocity–time graph gives a change in position. Components separate motion toward Earth from motion across its line of sight.

Read the physics: Signed area

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Read the vertical axis first. A flat line on a velocity–time graph means steady velocity. A flat line above zero means the object is moving.

02 HOW IT WORKS

Shade the region between velocity and the time axis. For steady velocity, multiply velocity by time. Add areas above zero; subtract the sizes of areas below zero. For distance, add all the sizes.

03 WHAT TO CHECK

A diagonal velocity has two components. For an angle measured from +x, use cosine for the x part and sine for the y part.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart moves at +3 m/s for 2 s , then −1 m/s for 2 s . Find its displacement and total distance. +4 m ; 8 m Return travel reduces Δ x , not distance.

Model check: Displacement = signed area under a velocity–time graph.

05 CONNECT TO XJ

For fictional XJ, the area of a velocity–time graph gives a change in position. Components separate motion toward Earth from motion across its line of sight.

SAY IT BEFORE YOU WRITE

Which shaded part subtracts from displacement? The area is ___ because it is ___ zero.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Signed area

ES: área con signo / PT: área com sinal

Meaning in my words	How can you tell?
How would you explain signed area to a friend? In my words, this means ___.	What do you do with the area below the time axis when you find displacement? I can tell because ___.
Example or sketch	**Non-example and why**
Sketch a rectangle at −2 m/s lasting 3 s. What displacement does it show? My example shows ___.	Someone labels the area −2 m/s. Which graph reading did they use? I would change ___ because ___.

Distance

ES: distancia / PT: distância

Meaning in my words	How can you tell?
How would you explain distance to a friend? In my words, this means ___.	A cart goes 3 m right and 2 m left. How much ground do its wheels cover? I can tell because ___.
Example or sketch	**Non-example and why**
Draw a trip that covers 6 m and finishes at its starting point. My example shows ___.	A partner writes −5 m for distance. What does the minus sign tell you? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Component

ES: componente / PT: componente

Meaning in my words	How can you tell?
How would you explain component to a friend? In my words, this means ___.	If east and north are positive, which component is negative for a southeast trip? I can tell because ___.
Example or sketch	**Non-example and why**
What two arrows could you use to build a displacement of 3 m east and 4 m north? My example shows ___.	A classmate counts the diagonal and both parts as three separate trips. What is counted twice? I would change ___ because ___.

Angle θ

ES: ángulo / PT: ângulo

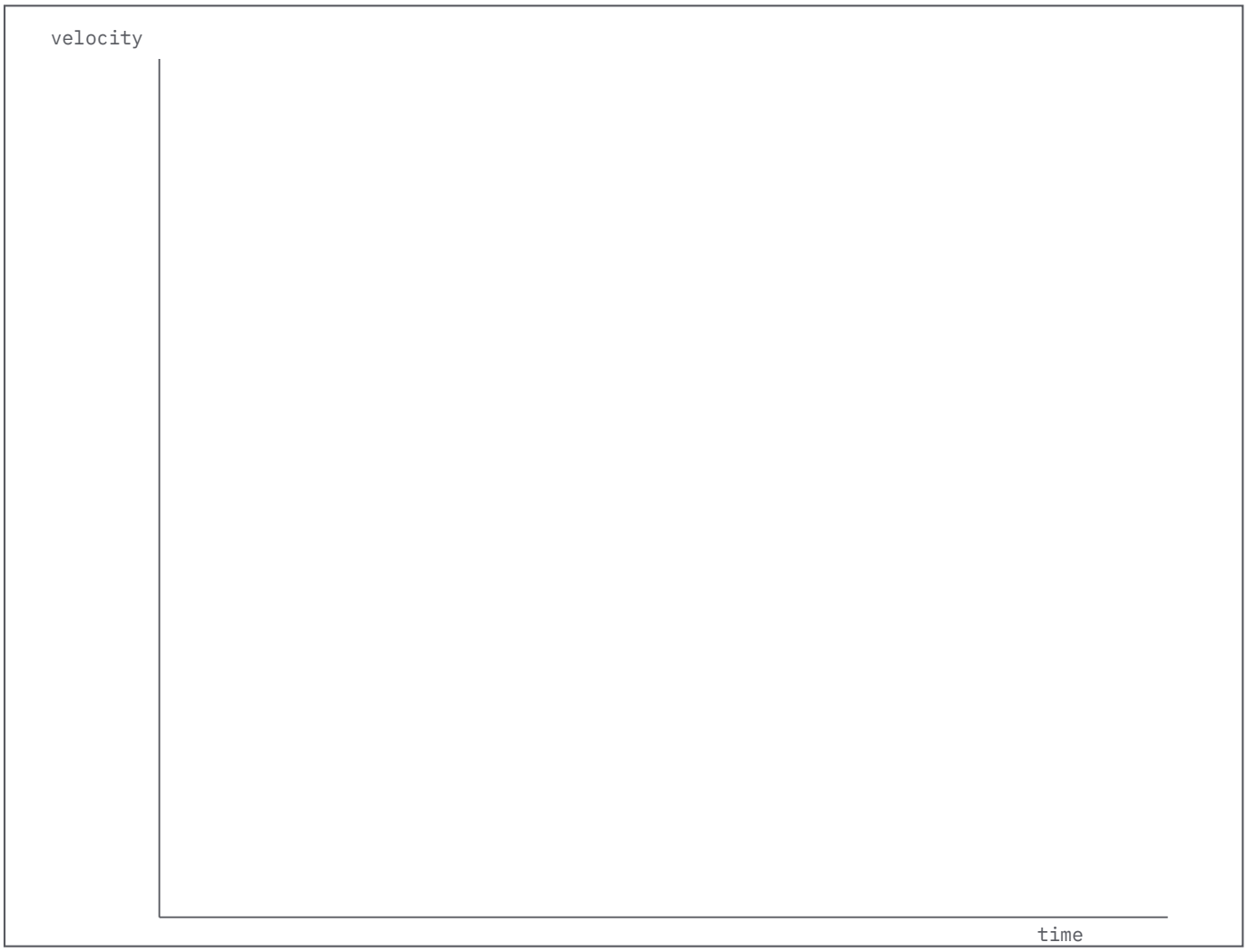
Meaning in my words	How can you tell?
When you label an arrow with an angle, what does that angle tell someone? In my words, this means ___.	What extra information do you need when a vector is labeled “30 degrees”? I can tell because ___.
Example or sketch	**Non-example and why**
Show 30 degrees north of east on a sketch. My example shows ___.	Does 30 degrees east of north point the same way as 30 degrees north of east? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw and shade opposite-side rectangles. Put m/s on the vertical and s on the horizontal axis.

Labels you can use: above · below · area.



Tell a partner

The area is ___ because it is ___ zero.

Where does the camera finish?

04

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use signed graph areas to find displacement.

CHOOSE ONE JOB TOGETHER

TRACKING ENGINEER ☐

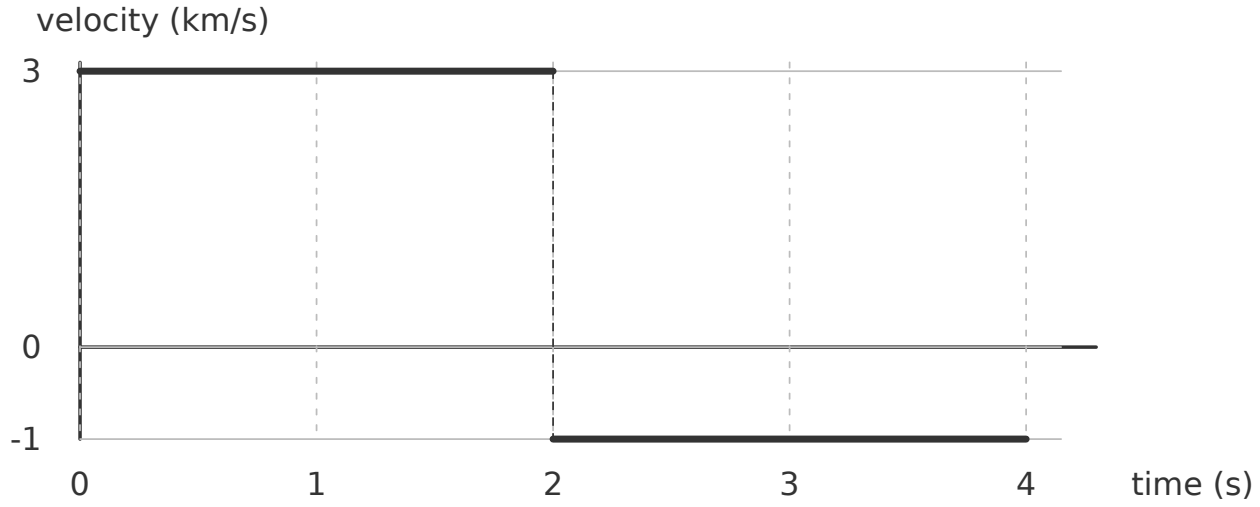
Shade both parts of the velocity graph. Radio the camera's final change in position, including direction.

REPLAY PRODUCER ☐

Mark the outward and return shots. Report total distance and net movement to the editor.

Word help: area / return / displacement

A moving camera follows a model at **+3 km/s** for **2 s**, then **–1 km/s** for **2 s**. The editor needs the end location and the total footage distance.



01 OUR FIRST CALL

Where does the camera finish relative to its start? How far did it travel?

The camera travels ___ but ends ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the new caption. Which number belongs on the location graphic? Explain.

Now we would say ___ because ___.

Switch speakers. If the camera also moves sideways, why does that motion need its own velocity-time graph?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks